SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE CODE: CSA1610
COURSE NAME: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO5: Apply association rule mining techniques to discover interesting relationships and patterns within large datasets

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QUESTIONS: 05 **Total Marks:50**

Annexure A

Pseudocode Writing Task (Algorithm Design)

Criteria	Problem Understanding ***(2)***	Algorithm Design ***(3)***	Use of Control Structures ***(2)***	Pseudocode Structure ***(1)***	Completeness ***(2)***	***Total(10)***
Weightage	*20%*	30%	20%	10%	20%	*100%*
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I Tejaswini.M (192411079) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Tejaswini.M

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation
Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach
Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
Pseudocode Structure	10%	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear
Completeness	20%	Handles all cases; optimized approach	Handles most cases; reasonable efficiency	Handles basic cases only	Incomplete or inefficient

Annexure A

Q1. Dataset: Online Shopping Transactions

Order ID	Products Purchased
O1	Mobile, Earphones
O2	Mobile, Charger
O3	Earphones, Charger
O4	Mobile, Earphones, Charger
O5	Mobile, Earphones

Question:
Design pseudocode for the **Apriori algorithm** to discover frequent product combinations from the online shopping dataset using a specified minimum support threshold. Clearly illustrate the candidate itemset generation and pruning process at each iteration.

Step 1 – Count individual items

Item	Count
Mobile	4
Earphones	3
Charger	2

All three satisfy the minimum count of 2.
So:

$$L_1 = \{M, o, bileEarphonesCharger\}$$

Step 2 – Generate candidate 2-itemsets
Candidates:

- {Mobile, Earphones}
- {Mobile, Charger}
- {Earphones, Charger}

Counts:

Candidate	Count	Support
Mobile, Earphones	3	60%
Mobile, Charger	2	40%
Earphones, Charger	2	40%

All are frequent.

$$L_2 = \{\{M, obileEarphones\}, \{M, obileCharger\}, \{E, arphonesCharger\}\}$$

Step 3 – Generate candidate 3-itemset
Candidate:
{Mobile, Earphones, Charger}
It occurs only in O4.

$$Support = \frac{1}{5} = 20\%$$

Since 20% < 40%, it is pruned.
Therefore:

$$L_3 = \emptyset$$

Pseudocode
APRIORI(Transactions, MinimumSupport)

```
Input:
   Transactions
   MinimumSupport

Output:
   Frequent itemsets

Begin

   L1 ← Find frequent 1-itemsets

   k ← 2

   While L(k-1) is not empty do

      Ck ← GenerateCandidates(L(k-1))

      For each transaction T do
         For each candidate c in Ck do
            If c is contained in T then
               Increase count(c)
            End If
         End For
      End For

      Lk ← empty set

      For each candidate c in Ck do
         support ← count(c) / NumberOfTransactions

         If support >= MinimumSupport then
            Add c to Lk
         Else
            Prune c
         End If
      End For

      k ← k + 1

   End While

   Return all Lk
```

End
Final Answer
Frequent combinations are:

- **{Mobile, Earphones} – 60%**
- **{Mobile, Charger} – 40%**
- **{Earphones, Charger} – 40%**

The 3-itemset is removed because its support is only **20%**.

Q2. Dataset: Library Book Borrowing Records

Borrow ID	Books Borrowed
B1	Data Science, Python
B2	Python, Machine Learning
B3	Data Science, Machine Learning
B4	Data Science, Python, Machine Learning
B5	Python, Machine Learning

Question:
Develop pseudocode for the **FP-Growth algorithm** to identify frequent borrowing patterns without generating candidate itemsets. Explain the steps involved in constructing the FP-Tree using the given borrowing records. *(BL4)*

Step 1 – Count item frequencies

Item	Count
Python	4
Machine Learning	4
Data Science	3

All three are frequent.
Step 2 – Sort items by frequency
Descending order:

1. Python – 4
2. Machine Learning – 4
3. Data Science – 3

For equal frequency, either consistent ordering can be used.
Step 3 – Ordered transactions
For example:
B1
Data Science, Python
becomes:
Python → Data Science
B2
Python, Machine Learning
becomes:
Python → Machine Learning
B3
Data Science, Machine Learning

becomes:
Machine Learning → Data Science
B4
Data Science, Python, Machine Learning
becomes:
Python → Machine Learning → Data Science
B5
Python, Machine Learning
becomes:
Python → Machine Learning
FP-Tree idea
The common prefix **Python → Machine Learning** is shared by B2, B4 and B5.
This means FP-Growth stores the common pattern compactly instead of generating every candidate combination.
Pseudocode

```
FP_GROWTH(Transactions, MinimumSupport)

Begin

   Count frequency of every item

   Remove items whose support is
   less than MinimumSupport

   Sort remaining items by descending frequency

   Create an empty FP-Tree

   For each transaction T do

      Remove infrequent items from T

      Sort items in T according to frequency

      Insert ordered T into FP-Tree

   End For

   Create Header Table

   For each item in Header Table do

      Construct conditional pattern base

      Construct conditional FP-Tree

      If conditional tree contains a single path then
         Generate all combinations from the path
      Else
         Recursively apply FP_GROWTH
```

End If

End For

Return frequent patterns

End

Why FP-Growth is useful

Apriori repeatedly generates candidate itemsets.

FP-Growth avoids this candidate-generation process and instead:

1. Counts frequent items.
2. Builds an FP-Tree.
3. Creates conditional pattern bases.
4. Builds conditional FP-Trees.
5. Recursively discovers frequent patterns.

Q3. Dataset: Frequently Purchased Services

Service Combination	Support Count
{Internet}	4
{Streaming}	4
{Cloud Storage}	3
{Internet, Streaming}	3
{Streaming, Cloud Storage}	2

Question:

Write pseudocode to generate **association rules** from the given frequent service combinations. Include the procedures for confidence calculation, rule generation, and filtering based on a minimum confidence threshold. *(BL3)*

Rule 1

$$Internet \rightarrow Streaming$$

Formula:

$$Confidence(A \rightarrow B) = \frac{Support(A \cup B)}{Support(A)} \times 100$$

Therefore:

$$Confidence = \frac{3}{4} \times 100 = 75\%$$

Since:

$$75\% \geq 70\%$$

✅ Accept.

Rule 2

$$Streaming \rightarrow Internet$$

$$Confidence = \frac{3}{4} \times 100 = 75\%$$

✅ Accept.

Rule 3

$$Streaming \rightarrow Cloud\ Storage$$
$$Confidence = \frac{2}{4} \times 100 = 50\%$$

❌ Reject.

Rule 4

$$Cloud\ Storage \rightarrow Streaming$$
$$Confidence = \frac{2}{3} \times 100 = 66.67\%$$

❌ Reject.

Pseudocode

```
GENERATE_RULES(FrequentItemsets, MinimumConfidence)

Begin

  For each frequent itemset I do

    Generate all non-empty subsets A of I

    For each subset A do

      B ← I - A

      confidence ←
        Support(I) / Support(A)

      confidence ← confidence × 100

      If confidence >= MinimumConfidence then

        Generate rule:
        A → B

        Output rule and confidence

      Else

        Reject rule

      End If

    End For

  End For
```

End

Final Accepted Rules

Rule	Confidence	Result
Internet → Streaming	75%	✅ Accept
Streaming → Internet	75%	✅ Accept
Streaming → Cloud Storage	50%	❌ Reject
Cloud Storage → Streaming	66.67%	❌ Reject

Q4. Dataset: Student Course Enrollment

Student ID	Courses Enrolled
S1	Python, Database
S2	Python, AI
S3	Database, AI
S4	Python, Database, AI
S5	Python

Question:
Design pseudocode for **constraint-based frequent itemset mining** where only itemsets containing the course **"Python"**are considered during the mining process. Explain how the constraint affects candidate generation and pruning. *(BL4)*

Step 1 – Apply the constraint
The constraint is:

$$Python \in Itemset$$

So we only consider:

- {Python}
- {Python, Database}
- {Python, AI}
- {Python, Database, AI}

We do **not** generate:

- {Database}
- {AI}
- {Database, AI}

because they do not contain Python.

Step 2 – Calculate support

{Python}

Occurs in:
S1, S2, S4, S5
Count = 4

$$Support = 80\%$$

✅ Frequent.

{Python, Database}

Occurs in:
S1, S4
Count = 2

$$Support = 40\%$$

✅ Frequent.
{Python, AI}
Occurs in:
S2, S4
Count = 2

$$Support = 40\%$$

✅ Frequent.
{Python, Database, AI}
Occurs only in S4.
Count = 1

$$Support = 20\%$$

❌ Pruned.
Pseudocode

```
CONSTRAINT_BASED_MINING(Transactions, MinimumSupport)

Begin

   RequiredItem ← "Python"

   Generate candidate itemsets

   For each candidate C do

      If RequiredItem is not in C then
         Prune C
      Else

         Calculate support(C)

         If support(C) >= MinimumSupport then
            Add C to FrequentItemsets
         Else
            Prune C
         End If

      End If

   End For

   Return FrequentItemsets

End
```

Final Answer

Itemset	Support	Result
{Python}	80%	✅ Frequent
{Python, Database}	40%	✅ Frequent
{Python, AI}	40%	✅ Frequent
{Python, Database, AI}	20%	❌ Pruned

Q5. Dataset: Hospital Treatment Hierarchy

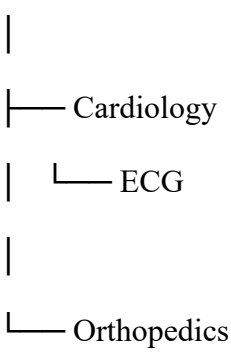
Patient ID	Treatments Received
P1	Healthcare, Cardiology, ECG
P2	Healthcare, Orthopedics
P3	Healthcare, Cardiology
P4	Healthcare, Cardiology, ECG
P5	Healthcare, Orthopedics

Question:
Develop pseudocode for **multilevel association rule mining** using the treatment hierarchy provided in the dataset. Explain how rules are generated at different hierarchy levels and how concept hierarchies improve pattern discovery.

Treatment hierarchy

A simple representation is:

Healthcare



Therefore, we have multiple levels:

Level 1:

Healthcare

Level 2:

Cardiology
Orthopedics

Level 3:

ECG

Level 1

Healthcare appears in all five transactions.

$$Support(Healthcare) = \frac{5}{5} = 100\%$$

Level 2

Cardiology

Occurs in P1, P3, P4.

$$Support = \frac{3}{5} = 60\%$$

Orthopedics

Occurs in P2, P5.

$$Support = \frac{2}{5} = 40\%$$

Level 3

ECG

Occurs in P1 and P4.

$$Support = \frac{2}{5} = 40\%$$

Example association

Consider:

$$Cardiology \rightarrow ECG$$

Cardiology occurs 3 times.

Both Cardiology and ECG occur together in P1 and P4.

Therefore:

$$Confidence = \frac{2}{3} \times 100 = 66.67\%$$

Reverse:

$$ECG \rightarrow Cardiology$$

Since both ECG records also contain Cardiology:

$$Confidence = \frac{2}{2} \times 100 = 100\%$$

Pseudocode

MULTILEVEL_ASSOCIATION_MINING

(Transactions, Hierarchy, MinimumSupport)

Begin

For each level in Hierarchy do

Identify items at current level

Count support of each item

Remove items below MinimumSupport

Generate frequent itemsets

For each frequent itemset do

Generate possible association rules

Calculate support

Calculate confidence

Store valid rules

End For

End For

Return all rules from all hierarchy levels

End

Final Results

Level	Item	Support
Level 1	Healthcare	100%
Level 2	Cardiology	60%
Level 2	Orthopedics	40%
Level 3	ECG	40%

Example rule:

Cardiology → ECG

- Support = **40%**
- Confidence = **66.67%**

ECG → Cardiology

- Support = **40%**
- Confidence = **100%**

How concept hierarchies improve pattern discovery

A hierarchy allows the mining algorithm to discover patterns at different levels of detail.

For example:

Healthcare

↓

Cardiology

↓

ECG

At a higher level, we discover broad treatment patterns. At lower levels, we discover more specific relationships.

This helps users move from **general patterns to specific patterns** and can reveal relationships that may not be visible when only one level is analyzed.